

# **SOFTWARE REQUIREMENTS SPECIFICATION**

*for*

## **Smart Complaint Management System**

Project ID: 67

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## 1. Introduction

### 1.1 Purpose

This Software Requirements Specification (SRS) document explains the requirements of the **Smart Complaint Management System**. The system is developed as a Software Engineering mini-project (Project ID 67) for the CSE (AI/ML) Department.

It describes the main features of the system, such as **submitting complaints, tracking complaints, resolving complaints, escalation, category and priority setting, SLA monitoring, and the admin dashboard**.

This document acts as the **main reference for what the system should do**. It focuses on the system requirements and does not explain how the system will be technically developed. The technical details are covered in the High-Level Architecture and Software Design documents.

### 1.2 Intended Audience and Reading Suggestions

This document is intended for the project team, faculty evaluators, developers, testers, project managers, and future maintainers. Readers should begin with the Introduction and Overall Description, then use the System Features and Nonfunctional Requirements for detailed requirements. The traceability matrix can be used to connect requirements with design, code, and testing.

### 1.3 Product Scope

The Smart Complaint Management System is a web-based platform that allows individuals (citizens, students, or employees, depending on the deployment context) to lodge complaints against a service, product, or process, and allows an organization to track, categorize, prioritize, assign, escalate, and resolve those complaints within defined Service Level Agreements (SLAs).

Key objectives of the product are:

- Provide a simple, guided interface for lodging a complaint with supporting evidence (text, attachments).
- Automatically or manually tag complaints with a category and priority.
- Track the full lifecycle of a complaint through well-defined statuses (e.g., New, In Progress, Escalated, Resolved, Closed, Reopened).
- Automatically escalate complaints that breach SLA thresholds to the next level of authority.

- Give administrators real-time visibility into complaint volume, ageing, and SLA compliance through a dashboard.
- Notify complainants and staff of status changes at each stage of the workflow.

The system reduces manual complaint-handling overhead, improves accountability through SLA monitoring, and provides data that supports continuous service improvement.

## 1.4 References

- Anurag Saha, “Automated Complaint Classification and Routing Using NLP and Machine Learning,” *International Journal of Innovative Research in Technology (IJIRT)*, vol. 11, no. 6, pp. 756–759, 2024.
- “An Intelligent Urban Civic Complaint Management System with AI-Based Classification and SLA Escalation,” *International Journal of Innovative Research in Technology (IJIRT)*, 2026.
- “NLP-Enhanced Intelligent Complaint Management System for Municipal Services,” *International Journal of Research in Applied Science & Engineering Technology (IJRASET)*.
- “Citizen Grievance Classification and Decision Support Using NLP,” *International Journal of Scientific Research and Engineering Development (IJSRED)*, vol. 9, no. 2.
- “Smart Complaint Analysis and Classification System Using Machine Learning and NLP,” *Zenodo*, Arab International University repository, 2026.
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## 2. Overall Description

### 2.1 Product Perspective

The Smart Complaint Management System is a new, self-contained web application. It is not a replacement for an existing system and has no dependency on a legacy platform. It consists of the following major components:

- A client-facing web front-end used by complainants, staff, and administrators.
- A back-end REST API layer that implements business logic — complaint lifecycle, escalation rules, SLA computation.
- A relational database that stores users, complaints, categories, SLA policies, and audit logs.
- A notification service (email/SMS/in-app) triggered by status and SLA events.
- A scheduled job (SLA monitor) that periodically evaluates open complaints against SLA deadlines and triggers escalation.

Figure 1 in Section 4 shows the major actors and the use cases they interact with.

### 2.2 Product Functions

At a high level, the system must let users perform the following functions (detailed in Section 5):

- Register, authenticate, and manage user profiles with role-based access (Complainant, Staff/Agent, Administrator).

- Lodge a new complaint with a description, category, attachments, and contact details.
- Automatically or manually assign a category and priority (Low/Medium/High/Critical) to a complaint.
- Track complaint status changes and view a complete history/timeline.
- Assign complaints to the appropriate staff member or department.
- Escalate complaints automatically when SLA deadlines are breached, or manually by staff/admin.
- Monitor SLA compliance in real time and flag at-risk or breached complaints.
- View an administrative dashboard with metrics — open/closed counts, average resolution time, SLA breach rate, category-wise distribution.
- Generate and export reports for a selected time range, category, or department.
- Send notifications to complainants and staff on status changes, assignment, and escalation.

## 2.3 User Classes and Characteristics

| User Class              | Characteristics                                                                                                                                            |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Complainant             | General public / employee / student who lodges complaints and tracks their own complaints. Minimal technical expertise assumed; needs a simple, guided UI. |
| Support Staff / Agent   | Handles assigned complaints, updates status, adds resolution notes. Moderate familiarity with the domain; uses the system daily.                           |
| Administrator           | Configures categories, SLA policies, and escalation rules; manages users and roles; views the dashboard and reports. Highest privilege level.              |
| Automated SLA Scheduler | Not a human user — a background system process that evaluates SLA timers and triggers escalation/notification events.                                      |

Administrators and Support Staff are the most important user classes to satisfy fully, since SLA monitoring, escalation, and dashboard accuracy are the core value proposition of the system.

## 2.4 Operating Environment

- Client: Any modern web browser (Chrome, Firefox, Edge, Safari) on desktop, tablet, or mobile, supporting a responsive layout.
- Server: Linux-based application server hosting the REST API (e.g., containerized with Docker).
- Database: Relational database (e.g., PostgreSQL/MySQL).
- Deployment: Cloud or on-premise infrastructure with CI/CD pipeline (Jenkins) and containerization (Docker), consistent with the course's mandated toolchain.

## 2.5 Design and Implementation Constraints

- Must follow Agile methodology with GitHub (or Jira) used for user stories and backlog tracking, as mandated by the course guidelines.
- Must have CI/CD pipeline, code reviews, and static/security validation (e.g., SonarQube) integrated into the workflow.
- All complaint data and personally identifiable information (PII) must be access-controlled by role.

- SLA thresholds and escalation rules must be configurable by the Administrator without a code change (data-driven, not hard-coded).
- The system must be modular so that notification channels (email/SMS/in-app) can be added or replaced independently.

## 2.6 Assumptions and Dependencies

- It is assumed that each complaint belongs to exactly one category at a time and is owned by exactly one assignee at a time.
- The system depends on a third-party email/SMS gateway (e.g., SMTP provider or SMS API) for outbound notifications.
- SLA durations are assumed to be defined per category/priority combination by the Administrator during system setup.
- It is assumed that team members have access to GitHub Education and standard open-source tooling for development and testing.

## 3. External Interface Requirements

### 3.1 User Interfaces

- A public-facing "Lodge Complaint" form with fields for description, category (optional — can be auto-suggested), attachments, and contact information.
- A "My Complaints" dashboard for complainants showing status, timeline, and any staff comments.
- A Staff console listing assigned complaints, sortable by priority, SLA remaining time, and status.
- An Administrator dashboard with charts/widgets for open/closed counts, SLA compliance %, and category-wise breakdown, plus screens to configure categories, priorities, and SLA rules.
- Consistent header/navigation, in-app notification bell, and standard error-message and confirmation-dialog patterns across all screens.

### 3.2 Software Interfaces

| Interface                   | Description                                                                                                                                |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Database (PostgreSQL/MySQL) | Stores users, complaints, categories, SLA rules, comments, attachments metadata, and audit trail via the back-end API's data-access layer. |
| Authentication provider     | Handles login/registration; may be a self-hosted JWT-based auth module or an external OAuth provider.                                      |
| Email/SMS gateway           | Sends outbound notifications for status changes, assignment, and escalation events, invoked by the notification service via REST/SMTP API. |
| File/Object storage         | Stores complaint attachments (images, PDFs) referenced by URL from the complaint record.                                                   |

### 3.3 Communications Interfaces

- All client–server communication uses HTTPS with a RESTful JSON API.

- Outbound email notifications use SMTP or a transactional email API (e.g., SendGrid); SMS notifications use a carrier/API gateway.
- Session/authentication tokens (JWT) are transmitted via secure, HttpOnly cookies or Authorization headers.

## 4. Analysis Models

Figure 1 below shows the use case diagram for the Smart Complaint Management System, identifying the primary actors (Complainant, Support Staff/Agent, Administrator, and the Automated SLA Scheduler) and the use cases each interacts with.

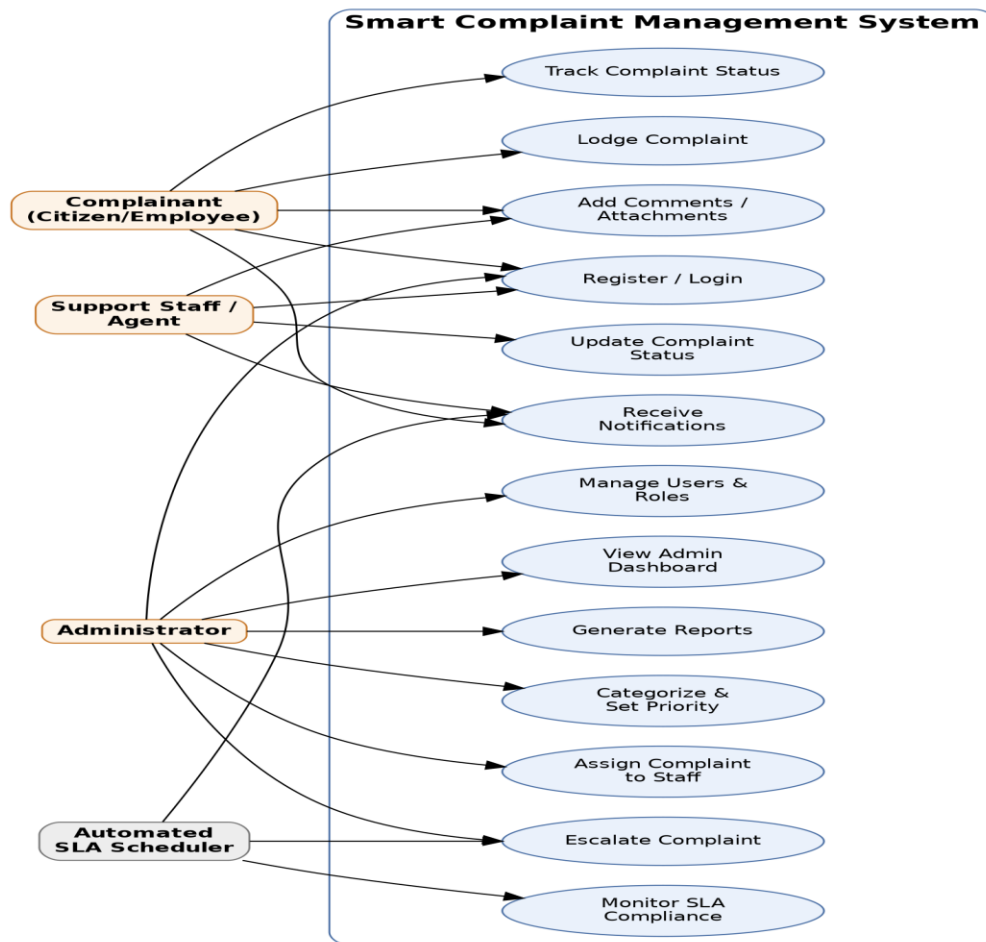


Figure 1: Use Case Diagram — Smart Complaint Management System

## 5. System Features

This section organizes the functional requirements of the product by system feature, the major services provided by the product.

### 5.1 User Registration and Role-Based Access

#### 5.1.1 Description and Priority

Allows Complainants, Support Staff, and Administrators to register/log in and be granted access according to their role. Priority: High.

### 5.1.2 Stimulus/Response Sequences

User submits registration details → system validates and creates account → user logs in with credentials → system authenticates and issues a session token scoped to the user's role.

### 5.1.3 Functional Requirements

**REQ-1:** The system shall allow a new user to register with name, email, phone number, and password.

**REQ-2:** The system shall validate email uniqueness and password strength during registration.

**REQ-3:** The system shall authenticate users via email/password and issue a session token (JWT) valid for a configurable duration.

**REQ-4:** The system shall enforce role-based access control (Complainant, Staff, Administrator) on every API endpoint and UI screen.

**REQ-5:** The system shall allow an Administrator to create, deactivate, or change the role of a user account.

## 5.2 Complaint Lodging

### 5.2.1 Description and Priority

Allows a Complainant to submit a new complaint with description, optional category, and attachments. Priority: High.

### 5.2.2 Stimulus/Response Sequences

Complainant fills the "Lodge Complaint" form and submits → system validates mandatory fields → system generates a unique complaint ID and sets status to "New" → system sends an acknowledgement notification to the complainant.

### 5.2.3 Functional Requirements

**REQ-6:** The system shall allow a Complainant to submit a complaint with a title, description (min. 20 characters), and optional category selection.

**REQ-7:** The system shall allow attachment of up to 5 files (images/PDF, max 10 MB each) per complaint.

**REQ-8:** The system shall auto-generate a unique, human-readable complaint reference ID (e.g., CMP-2026-000123) upon successful submission.

**REQ-9:** The system shall set the initial status of a newly lodged complaint to "New" and record the submission timestamp.

**REQ-10:** The system shall reject submission and display a validation error if mandatory fields are missing or invalid.

**REQ-11:** The system shall send an acknowledgement notification (in-app and email) to the complainant upon successful lodging, including the complaint reference ID.

## 5.3 Category Tagging and Priority Setting



### 5.3.1 Description and Priority

Allows a category and priority level to be assigned to a complaint, either automatically (rule/keyword-based suggestion) or manually by staff/admin. Priority: High.

### 5.3.2 Stimulus/Response Sequences

Complaint is submitted or reviewed → system suggests a category based on keywords, or staff selects a category manually → staff/system sets priority (Low/Medium/High/Critical) → SLA timer is initialized based on category+priority.

### 5.3.3 Functional Requirements

**REQ-12:** The system shall maintain a configurable list of complaint categories (e.g., Billing, Technical, Service Quality, Staff Behaviour) managed by the Administrator.

**REQ-13:** The system shall suggest a category automatically based on keyword matching in the complaint description, editable by staff before confirmation.

**REQ-14:** The system shall allow Staff/Administrator to set or change the priority of a complaint to one of: Low, Medium, High, Critical.

**REQ-15:** The system shall associate an SLA duration with each category–priority combination, configurable by the Administrator.

**REQ-16:** The system shall recompute the SLA deadline whenever the category or priority of a complaint is changed.

## 5.4 Complaint Tracking and Status Updates

### 5.4.1 Description and Priority

Allows complainants and staff to view the current status and full history of a complaint. Priority: High.

### 5.4.2 Stimulus/Response Sequences

Staff updates the status of an assigned complaint (e.g., New → In Progress → Resolved → Closed) → system timestamps the change and appends it to the complaint's audit trail → system notifies the complainant of the change.

### 5.4.3 Functional Requirements

**REQ-17:** The system shall support the following complaint statuses: New, Assigned, In Progress, Escalated, Resolved, Closed, Reopened.

**REQ-18:** The system shall allow only valid status transitions, enforced by a status transition rule set (e.g., Closed complaints can only move to Reopened).

**REQ-19:** The system shall log every status change with timestamp, actor (user), and optional remarks in an immutable audit trail.

**REQ-20:** The system shall allow a Complainant to view the real-time status and full history/timeline of their own complaints only.

**REQ-21:** The system shall allow a Complainant to reopen a Closed complaint within a configurable grace period (e.g., 7 days) with a justification.

## 5.5 Complaint Assignment

### 5.5.1 Description and Priority

Allows an Administrator (or an auto-assignment rule) to assign a complaint to a specific staff member or department. Priority: Medium.

### 5.5.2 Stimulus/Response Sequences

New complaint is categorized → system suggests an assignee based on category/department mapping, or Administrator assigns manually → assignee is notified → status changes to "Assigned".

### 5.5.3 Functional Requirements

**REQ-22:** The system shall allow an Administrator to manually assign a complaint to a staff member or department.

**REQ-23:** The system shall support an optional auto-assignment rule based on category-to-department mapping and current staff workload.

**REQ-24:** The system shall notify the assigned staff member immediately upon assignment or re-assignment.

**REQ-25:** The system shall allow reassignment of a complaint with a mandatory reason logged in the audit trail.

## 5.6 Escalation Workflow

### 5.6.1 Description and Priority

Automatically escalates complaints that are at risk of, or have breached, their SLA deadline to the next level of authority; also supports manual escalation. Priority: High.

### 5.6.2 Stimulus/Response Sequences

The Automated SLA Scheduler evaluates open complaints periodically → for complaints nearing or past their SLA deadline, the system escalates to the next-level assignee, changes status to "Escalated", and sends notifications to the new assignee, original assignee, and Administrator.

### 5.6.3 Functional Requirements

**REQ-26:** The system shall define an escalation hierarchy (e.g., Agent → Team Lead → Department Head → Administrator) configurable per category or department.

**REQ-27:** The system shall automatically escalate a complaint when its SLA deadline is breached without resolution.

**REQ-28:** The system shall send a pre-breach warning notification (e.g., at 80% of SLA duration elapsed) to the current assignee before automatic escalation occurs.

**REQ-29:** The system shall allow Staff/Administrator to manually escalate a complaint at any time, with a mandatory reason.

**REQ-30:** The system shall record every escalation event (trigger type, from/to assignee, timestamp) in the audit trail.

## 5.7 SLA Monitoring

### 5.7.1 Description and Priority

Continuously tracks each open complaint against its SLA deadline and surfaces at-risk and breached complaints. Priority: High.

### 5.7.2 Stimulus/Response Sequences

SLA Scheduler runs on a fixed interval (e.g., every 15 minutes) → evaluates remaining time for each open complaint → updates an SLA status flag (On Track / At Risk / Breached) → triggers notifications/escalation as configured.

### 5.7.3 Functional Requirements

**REQ-31:** The system shall compute and display the SLA deadline and remaining time for every open complaint.

**REQ-32:** The system shall classify each open complaint's SLA status as On Track, At Risk, or Breached, based on configurable thresholds.

**REQ-33:** The system shall exclude configured non-working hours/holidays from SLA countdown calculations, if business-hours SLA mode is enabled.

**REQ-34:** The system shall provide an SLA compliance report showing percentage of complaints resolved within SLA, filterable by category, department, and date range.

## 5.8 Admin Dashboard and Reporting

### 5.8.1 Description and Priority

Provides the Administrator with a real-time overview of complaint volumes, SLA compliance, and staff performance, and allows report export. Priority: Medium.

### 5.8.2 Stimulus/Response Sequences

Administrator logs in and opens the dashboard → system queries aggregated complaint data → displays widgets/charts → Administrator applies filters or exports a report (CSV/PDF).

### 5.8.3 Functional Requirements

**REQ-35:** The system shall display, on the Administrator dashboard, counts of complaints by status (New, In Progress, Escalated, Resolved, Closed).

**REQ-36:** The system shall display average resolution time and SLA breach rate, filterable by category, department, and date range.

**REQ-37:** The system shall display category-wise and priority-wise distribution of complaints as charts.

**REQ-38:** The system shall allow the Administrator to export dashboard data and complaint lists as CSV or PDF reports.

## 5.9 Notifications

### 5.9.1 Description and Priority

Sends in-app, email, and/or SMS notifications to relevant users on key lifecycle events. Priority: Medium.

### 5.9.2 Stimulus/Response Sequences

A trigger event occurs (submission, assignment, status change, SLA warning, escalation, resolution) → notification service composes a message from a template → delivers it via the user's preferred channel(s).

### 5.9.3 Functional Requirements

**REQ-39:** The system shall notify the complainant on: submission acknowledgement, status change, and resolution.

**REQ-40:** The system shall notify staff on: new assignment, SLA at-risk warning, and escalation.

**REQ-41:** The system shall allow users to configure their preferred notification channel(s) — in-app, email, and/or SMS — in their profile settings.

**REQ-42:** The system shall retain a log of all notifications sent, including delivery status, for audit purposes.

## 6. Other Nonfunctional Requirements

### 6.1 Performance Requirements

- The system shall support at least 500 concurrent users without response time degrading beyond 3 seconds for standard read operations.
- The complaint lodging form shall submit and confirm within 2 seconds under normal load.
- The SLA Scheduler job shall complete a full evaluation cycle over all open complaints within 5 minutes for up to 50,000 open complaints.

### 6.2 Safety Requirements

- The system shall never permanently delete a complaint or its audit trail; only soft-deletion/archival is permitted, to preserve accountability.
- Automatic escalation actions shall always be reversible by an Administrator, with the reversal itself logged.

### 6.3 Security Requirements

- All passwords shall be stored using a strong one-way hash (e.g., bcrypt/Argon2); plaintext passwords shall never be stored or logged.
- All data in transit shall be encrypted using TLS 1.2 or higher.
- The system shall enforce role-based access control such that a Complainant can view only their own complaints.
- All administrative actions (user management, SLA configuration, manual escalation) shall be logged with actor identity and timestamp.
- The system shall implement protection against common web vulnerabilities (SQL injection, XSS, CSRF), validated via static/security analysis (e.g., SonarQube) as part of CI/CD.

### 6.4 Software Quality Attributes

- **Availability:** The system shall target 99% uptime during business hours.
- **Usability:** A first-time Complainant shall be able to lodge a complaint without training, in under 5 steps.
- **Maintainability:** The codebase shall follow modular design principles (separate modules for auth, complaints, SLA, notifications) to allow independent development and testing.
- **Reliability:** No complaint or status-change event shall be lost even if the notification service is temporarily unavailable (events shall be queued and retried).

- Portability: The web client shall function correctly on the latest two major versions of Chrome, Firefox, and Edge.

## 6.5 Business Rules

- Only Administrators may create, edit, or deactivate complaint categories and SLA policies.
- Only Staff assigned to a complaint, or an Administrator, may change its status or add resolution notes.
- A complaint may be reopened only once within the grace period; subsequent reopen requests require Administrator approval.
- SLA timers pause automatically while a complaint is in a "Pending Complainant Response" state, if that status is used.

## 7. Other Requirements

- Database requirement: All complaint, user, and audit data shall be retained for a minimum of 3 years for compliance and reporting purposes.
- Legal/compliance: The system shall comply with applicable data-protection norms for storage of complainant personal data (e.g., contact information).
- Internationalization: The UI text shall be externalized into resource files to allow future localization, even though only English is required for the initial release.
- Reuse objective: Authentication and notification modules shall be built as reusable services that could be extracted for use in other Jackfruit mini-projects.

## Appendix A: Glossary

| Term                          | Definition                                                                                                                  |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Complaint                     | A formal record submitted by a Complainant describing an issue, along with metadata (category, priority, status, assignee). |
| SLA (Service Level Agreement) | The maximum allowed time within which a complaint of a given category/priority must be resolved.                            |
| Escalation                    | The act of forwarding a complaint to a higher level of authority, either automatically (SLA breach) or manually.            |
| Category                      | A classification label applied to a complaint (e.g., Billing, Technical) used for routing and reporting.                    |
| Priority                      | An urgency level (Low, Medium, High, Critical) assigned to a complaint, influencing its SLA duration.                       |
| Complainant                   | The end user who lodges a complaint.                                                                                        |
| Agent / Support Staff         | A user responsible for investigating and resolving assigned complaints.                                                     |
| Audit Trail                   | An immutable, chronological log of all actions taken on a complaint.                                                        |

## Appendix B: Field Layouts

The following tables describe the field layouts and attributes for key entities in the system.

### B.1 Complaint Registration Fields

| Field           | Length | Data Type      | Description                                               | Mandatory |
|-----------------|--------|----------------|-----------------------------------------------------------|-----------|
| Complaint ID    | 16     | Alphanumeric   | System-generated unique reference (e.g., CMP-2026-000123) | Y         |
| Complainant ID  | 16     | Alphanumeric   | Foreign key to the User record of the submitter           | Y         |
| Title           | 100    | String         | Short summary of the complaint                            | Y         |
| Description     | 2000   | String         | Detailed description of the issue                         | Y         |
| Category        | 50     | String         | Complaint category selected/suggested                     | N         |
| Priority        | 10     | Enum           | Low / Medium / High / Critical                            | N         |
| Status          | 20     | Enum           | Current lifecycle status of the complaint                 | Y         |
| SLA Deadline    | 8      | DateTime       | Computed deadline for resolution                          | Y         |
| Assignee ID     | 16     | Alphanumeric   | Foreign key to the assigned Staff user                    | N         |
| Attachment URLs | 500    | String (array) | Links to uploaded evidence files                          | N         |
| Created At      | 8      | DateTime       | Timestamp of complaint submission                         | Y         |

### B.2 User Registration Fields

| Field         | Length | Data Type    | Description                                           | Mandatory |
|---------------|--------|--------------|-------------------------------------------------------|-----------|
| User ID       | 16     | Alphanumeric | System-generated unique identifier                    | Y         |
| Name          | 100    | String       | Full name of the user                                 | Y         |
| Email         | 100    | String       | Unique email address, used for login and notification | Y         |
| Phone Number  | 15     | String       | Contact number, used for SMS notification             | N         |
| Password Hash | 255    | String       | One-way hash of the user's password                   | Y         |
| Role          | 20     | Enum         | Complainant / Staff / Administrator                   | Y         |
| Department    | 50     | String       | Department the Staff/Admin user belongs to            | N         |

B.3 Sample Report Requirements

| Complaint Summary Report | SLA Compliance Report   |
|--------------------------|-------------------------|
| Complaint ID             | Category                |
| Title                    | Priority                |
| Category                 | Total Complaints        |
| Priority                 | Resolved Within SLA     |
| Status                   | SLA Breached Count      |
| Assignee                 | SLA Compliance %        |
| Created Date             | Average Resolution Time |
| Resolved Date            | Department              |

Appendix C: Requirement Traceability Matrix

The following matrix will be completed progressively as Architecture, Design, Code, and Test artifacts are produced in subsequent phases.

| Sl. No | Requirement ID | Brief Description               | Architecture Ref. | Design Ref. | Code File Ref. | Test Case ID | System Case |
|--------|----------------|---------------------------------|-------------------|-------------|----------------|--------------|-------------|
| 1      | REQ-6          | Lodge a new complaint           | TBD               | TBD         | TBD            | TBD          | TBD         |
| 2      | REQ-17         | Complaint status transitions    | TBD               | TBD         | TBD            | TBD          | TBD         |
| 3      | REQ-27         | Automatic SLA-breach escalation | TBD               | TBD         | TBD            | TBD          | TBD         |
| 4      | REQ-32         | SLA status classification       | TBD               | TBD         | TBD            | TBD          | TBD         |
| 5      | REQ-35         | Admin dashboard status counts   | TBD               | TBD         | TBD            | TBD          | TBD         |